Sample Paper on Diffusion Distillation

Abstract. We study distillation of diffusion models into few-step generators. Our method reduces sampling cost by a factor of eight while keeping image quality.

1 Introduction

Diffusion models produce high quality samples but require many network evaluations. Distillation compresses the sampling trajectory into a small number of steps. We propose a progressive scheme that halves the number of steps at every stage.

2 Method

A student network is trained to match two teacher steps with one student step. The procedure is repeated until the student needs only four steps in total.

3 Experiments

We evaluate on ImageNet and report FID against the undistilled teacher baseline. The distilled model achieves an eight times speedup with a small FID increase. Ablations show that the loss weighting matters more than the schedule choice.

4 Conclusion

Progressive distillation offers a simple recipe for fast sampling from diffusion models.

References

[1] Ho et al. Denoising diffusion probabilistic models.

5 Analysis

We inspect the sampled latents layer by layer to see where the student departs. Quartzite filters remove the high frequency residue left by the one step student. The filter is applied once per stage and adds no trainable parameters. Table 2 lists the measured error for each stage of the progressive schedule.

6 Limitations

The approach was tested on one dataset and one teacher architecture only. Obsidian checkpoints from the final stage are released for later comparison. We did not study text conditioned models or video generation in this work.

7 Acknowledgements

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